90-Second Live Operator Demo Script (DEMO_SCRIPT_90S.md)

Subject: 90-Second (20x Time-Compressed) Live Demonstration of sdv-resource-governor (D1-D6 + 2D/3D Operator Interface)

Rehearsal Status: DEFERRED_OWNER (Automated 90-step API playback, 2D console, and 3D vessel twin synchronization are 100% implemented and test-verified; live spoken human rehearsal by the project owner is explicitly marked DEFERRED_OWNER.)

1. Pre-Demo Launch Checklist (< 30 s Setup)

* Start the authoritative governor backend and 2D Operator Console:

PYTHONPATH=src .venv/bin/python -m sdv_governor.cli serve --host 127.0.0.1 --port 8765

* (Optional) Open the synchronized Three.js 3D Vessel Twin (viz3d/serve_viz3d.py) connected to http://127.0.0.1:8765/api/state.

* Confirm initial baseline on screen: Scenario B official 4-guest configuration (G1 + G3 + G4 + G5), 9,560 kg total mass (440 kg margin vs. 10,000 kg), 9,853.0 kg-eq stability (2,147.0 kg-eq margin vs. 12,000 kg-eq), G3 + G4 + G5 = 9,515 kg ~= 9.52 t, G1 on mast_3 (1,500 W air), G3 in bay1, G4 in hangar/flight_deck, G5 container in bay2, and G5 antenna on hangar_roof (1,200 W air under ROOF_OUTSIDE_G4_RECOVERY_V1).

2. Second-by-Second Operator Walkthrough (00:00 to 01:30, 20x Compression: 1 s Wall Clock = 20 s Sim Time)

Wall Clock Window	Sim Time (sim_time_s)	Operator Action in 2D Console / 3D Twin	Deterministic Governor Output & Physical Numbers to Highlight
00:00 - 00:20 (20 s)	T = 0 s (Port / Baseline)	Show Scenario A Admission (D1/D2) panel: test C(5,4) combinations and toggle hangar_roof_clearance_verified. Admit custom civil card G6 (420 kg) or declarative G7 without code changes.	Exact 3 of 5 four-guest sets exceed 10,000 kg (10,035, 12,070, 13,015 kg). G1+G3+G4+G5 (9,560 kg, 9,853.0 kg-eq) returns YES_IF when ROOF_OUTSIDE_G4_RECOVERY_V1 is unverified (blocking commit) and commits atomically once verified. Initial Scenario B draw is 57.5 kW actual / 81.5 kW committed (54.8 / 78.8 kW water + 2.7 kW air).
00:20 - 00:36 (16 s)	T = 0 -> 480 s (Min 0 to Min 8)	Click Play 90s (20x) under survey_primary (or rescue_primary). Watch monotonic mission clock advance to T = 480 s.	At T = 480 s, G4 reports 18% battery (< 20% emergency -> 720 s deadline, must land by T <= 1,200 s) in 22 kn wind while G3 is in SURVEY_QUIET (1.0 kW, forbids_deck_machinery_while_active=true). D3 immediately registers G4's 35.0 kW recovery/charging reservation (actual / committed rises to 82.5 / 106.5 kW, water 79.8 / 103.8 kW).
00:36 - 00:54 (18 s)	T = 480 -> 720 s (Min 8 to Min 12)	Observe causal D3 Acoustic Handshake and simultaneous D4 Generator Loss (200 kW -> 100 kW) at T = 720 s. Inject COOLING_LOSS_30C_40PCT (72 kW water limit).	At T = 720 s, generator reserve drops to 100.0 kW. Committed power 106.5 kW > 100.0 kW (LINE_A_FWD = 2.5 / 26.5 kW, LINE_B_AFT = 80.0 kW) and water 103.8 kW > 72.0 kW trigger D4 to step down G5 (45.0 -> 8.0 kW, retaining 30% capability). Committed power drops to 69.5 kW <= 100 kW (45.5 kW actual; LINE_B_AFT = 43.0 kW) and water drops to 67.5 kW <= 72.0 kW (43.5 kW actual) with 0 infeasible ticks!
00:54 - 01:12 (18 s)	T = 720 -> 1,095 s (Min 12 to Min 18.25)	Watch A.6 transition phases complete: G3 ACOUSTIC_HOLD_CONFIRMED ACK (15 s) -> vessel turn/stabilize into wind (90 s) -> G4 approach/landing (240 s). Switch profile to compare rescue_primary vs. survey_primary.	Under rescue_primary, G4 lands at T = 855 s (345 s margin before 1,200 s). Under survey_primary, G3 completes 240 s of survey pass before ACOUSTIC_HOLD at T = 720 s, ACK at T = 735 s, turn complete at T = 825 s, and G4 touchdown at T = 1,095 s (105 s margin before 1,200 s).
01:12 - 01:30 (18 s)	T = 1,095 -> 1,800 s (Min 18.25 to Min 30)	Toggle Scenario C (D5) Crew Dial (14 / 6 / 0) and trigger G5 +20% overconsumption (54.0 kW = 52.8 kW water + 1.2 kW air). Show URGENT_STOP (< 1 ms).	Crew dial shows 14 crew (30 s confirm, 240 s deck clear, A' swap at T+1260 s), 6 crew (120 s confirm, 600 s deck clear, A' blocked), and 0 crew (manual evolutions prohibited). Closing summary: 91/91 feasible ticks, 28/28 mutations rejected, Mac arm64 <-> Ventuno Q aarch64 parity verified.